addresses, based on rules in its service table managed by the blockchain. The SAL can be programmed through a user space control plane, acting on service-level events triggered by active sockets (e.g., a service instance automatically registers on binding a socket). This gives network programmers hooks for ensuring service resolution systems are up-to-date.

A BlockCloud token is the network's native token and is used to enable an incentive-driven marketplace for IoT services. There are four main participants in the network:

- *Service providers* publish services to the marketplace and are rewarded with BLOC tokens based on their level of contribution.
- *Service users* subscribe to services from the marketplace and will pay BLOC tokens to consume various IoT services.
- *Service miners* maintain the ledger and protect the network from attacks by utilising a CoDAG ledger framework to achieve consensus. The miner that generates a stable block will be rewarded with BLOC tokens. Mining rewards will be released at an annual rate of 5 per cent over six years.
- *Verifiers* ensure the quality of services provided and help the system decide whether to reward or penalise a service provider. They are rewarded with BLOC tokens for verifying services.

BlockCloud will leverage various mechanisms such as PoS to verify services and truthful continuous double auction (TCDA) to distribute services fairly.

CloudPoS (proof of stake)

CloudPoS (proof of stake) is a type of consensus algorithm by which a cryptocurrency blockchain network aims to achieve distributed consensus. In CloudPoS based cryptocurrencies, the creator of the next block is chosen by various combinations of random selection and wealth or age (i.e., the stake). In contrast, the algorithm of proof of work based cryptocurrencies, such as bitcoin, uses mining — that is, the solving of computationally intensive puzzles to validate transactions and create new blocks.

The CloudPoS consensus initially requires the validators to decide what amount of resources they want to stake towards the consensus process, which is driven by the fact that the chances of becoming a leader in the epoch increase as the amount of resources staked are increased.

Without loss of generality, we model a stake *functionf(.)* for a validator i that is dependent on its total allocated resource Ri, implicit greediness parameter γi, and current resource utilisation vector (Rui). The function results in the stake vector $Xi=<Xci,Xsi,Xdi>$ for validator i, based on which this amount of resources is sliced off from the total allocated resources. Then the leader gets elected stochastically based on the individual stakes $[X]=\{Xi:i=1,\cdots N\}$.

If the leader is unavailable or unreachable, then a new leader is elected. After a leader is successfully selected, its block is propagated to perform a consistency-check on the transactions and match the Merkle root of the block to ensure that the leader's working chain aligns with other validators' chains. Finally, the leader collects its reward that was allocated for that particular epoch. CloudPoS consensus is executed in several phases during a particular epoch to finalise a block of transactions to be included in the mainstream blockchain. The same steps are iterated after the length of the blockchain increments by one block.

The five phases that constitute one epoch of the consensus are:

- Stake determination
- Resource staking and confirmation
- Leader election
- Block replication and verification
- Reward distribution

Advantages of the BlockCloud platform

- **Cloud data provenance:** User operations are monitored in real-time to collect provenance data, to support access control policy enforcement and intrusion detection.
- **Proof of stake validation:** As opposed to the proof of work based provenance model, the consensus process of BlockCloud is driven by the staked resources of virtual machines (VMs) housed in a federated cloud computing environment. The presence of validating VMs provides supervisory control over the consensus process.
- **Tamper-proof environment:** The data provenance record is collected and then published to the blockchain network, which protects the provenance data. All data on the blockchain is shared among blockchain network nodes. BlockCloud builds a public time-stamped log of all user operations on cloud data without the need for a trusted third party. Every provenance entry is assigned a blockchain receipt for future validation.
- **Provenance data validation:** The data provenance record is published globally on the blockchain network, where a number of blockchain nodes provide confirmation for every block. ProvChain uses blockchain receipt to validate every provenance data entry.

BlockCloud and IoT solution implementation

For large enterprises, it is very important for IoT solutions to have proper connectivity, security and reliability during network communication. Since most of the IoT based services work on real-time